

MCAC0813: APPLIED DATA STRUCTURE AND APPLICATIONS LAB

Objective: To implement the various data structures and to demonstrate the various algorithms of data structure.

Credits: 02

L-T-P-J: 0-0-4-0

Module No.	Content	Lab Hours
I	- Write a program for Linear search. - Write a program for Binary search. - Write a program for Bubble sort. - Write a program for Merge sort. - Write a program for Insertion sort. - Write a program for Selection sort. - Write a program for Quick sort. - Write a program to insert the element at given position in an array and display that array. - Write a program to delete the element from given position in an array and display that array. - Write a program to demonstrate various operations (length, copy, append, compare) on strings. - Write a program to demonstrate various operations (reverse, extracting a substring from left, extracting a substring from right, extracting a substring from middle) on strings. - Write a program to demonstrate various operations (create, push, pop, overflow, underflow, peek, display) of STACK using array implementation. - Write a program to demonstrate various operations (create, enqueue, dequeue, overflow, underflow, peek, display) of QUEUE using array implementation. - Write a program to demonstrate various operations (create, enqueue, dequeue, overflow, underflow, peek, display) of CIRCULAR QUEUE using array implementation. - Write a program to solve the problem of Tower of Hanoi by using recursion. - Write a program to demonstrate various operations (create, Traversing, searching, inserting an element at beginning, at end, after a given element, deleting an element from beginning, from end, after a given element) of a linked list. - Write a program to demonstrate various operations (create, Traversing, searching, inserting an element at beginning, at end, after a given element, deleting an element from beginning, from end, after a given element) of a doubly linked list. - Write a program to demonstrate various operations (create, push, pop, overflow, underflow, peek, display) of STACK using linked list. - Write a program to demonstrate various operations (create, enqueue, dequeue, overflow, underflow, peek, display) of QUEUE using linked list. - Write a program for addition of polynomials. - Write a program to illustrate traversal of a graph using Breadth- first search. - Write a program to illustrate traversal of a graph using Depth- first search. - Write a program for Minimum Spanning Tree. - Write a program to find shortest path for given source and destination. - Write a program to find shortest path among all pair of vertices.	48

Outcome: After completion of course, student will be able to:

- ☐ CO1: Demonstrate the associated operations in linear data structure like stack, Queue and link list.
- ☐ CO2: Demonstrate the associated operations in Binary Search Tree and Dijkstra's Algorithm.
- ☐ CO3: Implementation the sorting algorithms like Selection Sort, Bubble Sort, Insertion Sort, Merge Sort, Quick Sort and Heap Sort.

Mapping of Course Outcomes (COs) with Program Outcomes (POs) and Program Specific Outcomes (PSOs):

COs	POs/PSOs
CO1	PO1/PSO1
CO2	PO4/PSO1,PSO3
CO3	PO2/PSO3,PSO4